

Product & Quotient Rule

Section 2.3

9-3-25

PRODUCT RULE

$$\frac{d}{dx} [f(x) \cdot g(x)] = (f'(x) \cdot g(x)) + (g'(x) \cdot f(x))$$

Example:

$$y = (3x - 2x^2)(5 + 4x)$$

$$y' = ?$$

2 options

(1) FOIL & Power Rule

(2) Product Rule

$$y = 15x + 2x^2 - 8x^3$$

$$y' = 15 + 4x - 24x^2$$

$$y' = ((3-4x)(5+4x)) + (4(3x-2x^2))$$

$$y' = 15 - 20x - 12x - 16x^2 + 12x - 8x^2$$

$$y' = 15 + 4x - 24x^2$$

Another Example:

$$y = x^4 \sin x$$

$$y' = ((4x^3)(\sin x)) + ((x^4)(\cos x))$$

$$y' = 4x^3(\sin x) + x^4(\cos x)$$

also works

$$x^3(4\sin x + x\cos x)$$

QUOTIENT RULE

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$$

Example 1:

$$y = \frac{\sin x}{x^3}$$

$$y' = ? \frac{(\cos(x)(x^3)) - (3x^2)(\sin x)}{x^6} =$$

$$y' = \frac{x^3 \cos x - 3x^2 \sin x}{x^6} = \frac{x^2(x \cos x - 3 \sin x)}{x^6} =$$

also works

$$\frac{x \cos x - 3 \sin x}{x^4}$$

Example 2:

$$f(x) = \frac{x^2 + 3}{x^3 - 4x}$$

$$f'(x) = ?$$

$$f'(x) = \frac{((2x)(x^3 - 4x)) - (3x^2 - 4)(x^2 + 3)}{(x^3 - 4x)^2}$$

Example 3:

$$f(x) = \frac{x^2 + 2}{\sqrt{x}}$$

$$f'(x) = \frac{((2x)(x^{1/4})) - ((\frac{1}{2}x^{-3/4})(x^2 + 2))}{x^{1/2}}$$

YOU COULD DO THIS....

BUT, it's probably better to do this

$$f(x) = \frac{x^2}{x^{1/4}} + \frac{2}{x^{1/4}} = x^{7/4} + 2x^{-1/4}$$

$$f'(x) = \frac{7}{4}x^{3/4} - \frac{1}{2}x^{-5/4}$$

$$f'(x) = \frac{7}{4}x^{3/4} - \frac{1}{2x^{5/4}}$$

Today's Tid-Bit

if quotient has constant in

numerator (Ex: $\frac{9}{5x^2}$), write

func. w/ negative exponent

and solve w/ power rule:

Ex:

$$f(x) = \frac{9}{5x^2} = \frac{9}{5}x^{-2}$$

$$f'(x) = -\frac{18}{5}x^{-3}$$

There's it folks.